

[Aim: 100/100 in Maths]

आरंभ

TRIGONOMETRY

Lecture #8

#(K3B) $\Rightarrow \sin + \cos \theta$ given Sq. both sides $\rightarrow \underline{\sin \cdot \cos}$ निकाल ली

#LP: If $\cos \theta + \sin \theta = \sqrt{2} \cos \theta$, show that $\cos \theta - \sin \theta = \sqrt{2} \sin \theta$.

given

$$C + S = \sqrt{2} C$$

(sq. both sides)

$$(C + S)^2 = (\sqrt{2} C)^2$$

$$C^2 + S^2 + 2CS = 2C^2$$

$$1 + 2CS = 2C^2$$

$$2CS = 2C^2 - 1$$

$$\cos \theta \cdot \sin \theta = \frac{2\cos^2 \theta - 1}{2}$$

$$\sec + \tan = \frac{1}{\sec - \tan}$$

$$\csc - \cot = \frac{1}{\csc + \cot}$$

$$\sin + \cos = \frac{1}{\sin - \cos}$$

$$(\cos \theta - \sin \theta)^2 = \cos^2 \theta + \sin^2 \theta - 2\cos \theta \cdot \sin \theta$$

$$(\cos \theta - \sin \theta)^2 = 1 - \frac{2 \times (2\cos^2 \theta - 1)}{2}$$

$$(\cos \theta - \sin \theta)^2 = 1 - (2\cos^2 \theta - 1)$$

$$(\cos \theta - \sin \theta)^2 = 1 - 2\cos^2 \theta + 1$$

$$(\cos \theta - \sin \theta)^2 = 2 - 2\cos^2 \theta$$

$$(\cos \theta - \sin \theta)^2 = 2(1 - \cos^2 \theta)$$

$$\cos \theta - \sin \theta = \sqrt{2} \sin \theta$$

$$\cos \theta - \sin \theta = \sqrt{2} \sin \theta$$

HP

#1P:-

$$\sin \theta + \cos \theta = x$$

$$\text{ii) } \sin \theta - \cos \theta = ?$$

sq. both sides

$$(\sin \theta + \cos \theta)^2 = x^2$$

$$\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta = x^2$$

$$1 + 2 \sin \theta \cos \theta = x^2$$

$$2 \sin \theta \cos \theta = x^2 - 1$$

$$\sin \theta \cos \theta = \frac{x^2 - 1}{2}$$

$$\text{Now, } (\sin \theta - \cos \theta)^2 = \sin^2 \theta + \cos^2 \theta - 2 \sin \theta \cos \theta$$

$$(\sin \theta - \cos \theta)^2 = 1 - 2 \left(\frac{x^2 - 1}{2} \right)$$

$$(\sin \theta - \cos \theta)^2 = 1 - x^2 + 1$$

$$(\sin \theta - \cos \theta)^2 = 2 - x^2$$

$$\sin \theta - \cos \theta = \sqrt{2 - x^2}$$

$$(ii) \sin^3 \theta + \cos^3 \theta = ?$$

$$(\sin \theta + \cos \theta) \left(\underbrace{\sin^2 \theta + \cos^2 \theta}_1 - \underbrace{\sin \theta \cdot \cos \theta}_{\checkmark} \right)$$

$$\Rightarrow (x) \left(1 - \frac{x^2 - 1}{2} \right)$$

$$x \left(\frac{2 - x^2 + 1}{2} \right)$$

$$x \frac{3 - x^2}{2}$$



$$a^3 + b^3 = (a+b)(a^2 + b^2 - ab)$$

∞^0
(11)

$$\sin^4 \theta + \cos^4 \theta$$

$$\Rightarrow (\underbrace{\sin^2 \theta}_a)^2 + (\underbrace{\cos^2 \theta}_b)^2$$

$$\Rightarrow (\underbrace{\sin^2 \theta + \cos^2 \theta}_1)^2 - 2 \sin^2 \theta \cos^2 \theta$$

$$\Rightarrow (1)^2 - 2(\underbrace{\sin \theta \cdot \cos \theta}_{\text{green wavy line}})^2$$

$$\Rightarrow 1 - 2\left(\frac{x^2 - 1}{2}\right)^2$$

$$\Rightarrow 1 - \cancel{2} \frac{(x^2 - 1)^2}{\cancel{2}}$$

$$\boxed{1 - \frac{(x^2 - 1)^2}{2}}$$



$a^x b^x = (ab)^x$

$$(a+b)^2 = \underbrace{a^2 + b^2} + 2ab$$

$$(a+b)^2 - 2ab = a^2 + b^2$$

$$(iv) \sin^6 \theta + \cos^6 \theta$$

$$\left(\overset{a}{\sin^2 \theta} \right)^3 + \left(\overset{b}{\cos^2 \theta} \right)^3$$

$$a^3 + b^3$$

$$(a+b)(a^2+b^2-ab)$$

$$\Rightarrow (\sin^2 \theta + \cos^2 \theta) \left((\sin^2 \theta)^2 + (\cos^2 \theta)^2 - \sin^2 \theta \cdot \cos^2 \theta \right)$$

$$\Rightarrow 1 \times \left(\sin^4 \theta + \cos^4 \theta - (\sin \theta \cdot \cos \theta)^2 \right)$$

$$\Rightarrow \left[1 - \frac{(\chi^2 - 1)^2}{2} \right] - \left(\frac{\chi^2 - 1}{2} \right)^2$$

$\sin \theta$
 $\cos \theta$

}

$$(a+b)^2 = a^2 + b^2 + 2ab$$

$$(a-b)^2 = a^2 + b^2 - 2ab$$

use

#LP: If $\frac{x'}{a'} \cos \theta + \frac{y'}{b'} \sin \theta = 1$ and $\frac{x'}{a'} \sin \theta - \frac{y'}{b'} \cos \theta = 1$, prove that $\frac{x'^2}{a'^2} + \frac{y'^2}{b'^2} = 2$.

Squaring both sides

$$\textcircled{\text{I}} \Rightarrow \frac{x'^2}{a'^2} \cos^2 \theta + \frac{y'^2}{b'^2} \sin^2 \theta + 2 \frac{x'}{a'} \cos \theta \times \frac{y'}{b'} \sin \theta = 1$$

$$\textcircled{\text{II}} \Rightarrow \frac{x'^2}{a'^2} \sin^2 \theta + \frac{y'^2}{b'^2} \cos^2 \theta - 2 \frac{x'}{a'} \sin \theta \frac{y'}{b'} \cos \theta = 1$$

$$\frac{x'^2}{a'^2} (\cos^2 \theta + \sin^2 \theta) + \frac{y'^2}{b'^2} (\sin^2 \theta + \cos^2 \theta) = 2$$

$$\boxed{\frac{x'^2}{a'^2} + \frac{y'^2}{b'^2} = 2} \quad \text{H.P.}$$

#LP: If $a \sec \theta + b \tan \theta = m$ and $b \sec \theta + a \tan \theta = n$, prove that $a^2 + n^2 = b^2 + m^2$

Sq. both sides

$$\textcircled{I} \Rightarrow a^2 \sec^2 \theta + b^2 \tan^2 \theta + 2a \sec \theta b \tan \theta = m^2$$

$$\textcircled{II} \Rightarrow b^2 \sec^2 \theta + a^2 \tan^2 \theta + 2b \sec \theta a \tan \theta = n^2$$

$$\underline{a^2 \sec^2 \theta - b^2 \sec^2 \theta + b^2 \tan^2 \theta - a^2 \tan^2 \theta = m^2 - n^2}$$

$$a^2 (\sec^2 \theta - \tan^2 \theta) - b^2 (\sec^2 \theta - \tan^2 \theta) = m^2 - n^2$$

$$a^2 - b^2 = m^2 - n^2$$

$$\boxed{a^2 + n^2 = m^2 + b^2}$$

HP

#LP: If $x = r \sin \alpha \cos \beta$, $y = r \sin \alpha \sin \beta$ and $z = r \cos \alpha$ prove that: $x^2 + y^2 + z^2 = r^2$.

LHS

$$\left[\underbrace{x^2}_{r^2 \sin^2 \alpha \cos^2 \beta} + \underbrace{y^2}_{r^2 \sin^2 \alpha \sin^2 \beta} \right] + z^2$$

$$\underline{\underline{r^2 \sin^2 \alpha (\cos^2 \beta + \sin^2 \beta)}} + \underline{\underline{r^2 \cos^2 \alpha}}$$

$$\rightarrow \underline{\underline{r^2 \sin^2 \alpha}} + \underline{\underline{r^2 \cos^2 \alpha}}$$

$$\rightarrow r^2 (\underline{\underline{\sin^2 \alpha + \cos^2 \alpha}})$$

$\rightarrow$

$$r^2 = \text{RHS}$$



#LP: If $a \cos \theta - b \sin \theta = c$, prove that $a \sin \theta + b \cos \theta = \sqrt{a^2 + b^2 - c^2}$.

Sq. both
sides

$$(a \cos \theta - b \sin \theta)^2 = c^2$$

$$a^2 \cos^2 \theta + b^2 \sin^2 \theta - 2a \cos \theta \cdot b \sin \theta = c^2$$

$$\Rightarrow a^2 (1 - \sin^2 \theta) + b^2 (1 - \cos^2 \theta) - 2a \cos \theta \cdot b \sin \theta = c^2$$

$$\Rightarrow \underline{a^2} - \underline{a^2 \sin^2 \theta} + \underline{b^2} - \underline{b^2 \cos^2 \theta} - \underline{2a \cos \theta \cdot b \sin \theta} = \underline{c^2}$$

$$\Rightarrow a^2 + b^2 - c^2 = \cancel{a^2 \sin^2 \theta} + \cancel{b^2 \cos^2 \theta} + 2 \cancel{a \cos \theta} \cdot \cancel{b \sin \theta}$$

$A^2 + B^2 + 2AB$
 $(A+B)^2$

$$a^2 + b^2 - c^2 = (a \sin \theta + b \cos \theta)^2$$

~~HP~~

$$\sqrt{a^2 + b^2 - c^2} = a \sin \theta + b \cos \theta$$

#LP: If $\tan\theta + \sin\theta = m$, $\tan\theta - \sin\theta = n$, show that $m^2 - n^2 = 4\sqrt{mn}$.

LHS

$$m^2 - n^2$$

$$(m+n)(m-n)$$

$$(t + \cancel{s} + t - \cancel{s})(t + \cancel{s} - t + \cancel{s})$$

$$\Rightarrow (2t)(2s)$$

$$\boxed{\text{LHS} \Rightarrow 4 \tan\theta \sin\theta}$$

RHS

$$4\sqrt{mn}$$

$$\Rightarrow 4\sqrt{(t+s)(t-s)}$$

$$\Rightarrow 4\sqrt{t^2 - s^2}$$

$$\Rightarrow 4\sqrt{\frac{s^2}{c^2} - s^2}$$

$$\Rightarrow 4 \cdot \sqrt{s^2 \left(\frac{1}{c^2} - 1 \right)}$$

$$\Rightarrow 4s \sqrt{\frac{1-c^2}{c^2}}$$

$$\Rightarrow 4s \cdot \sqrt{\frac{s^2}{c^2}}$$

$$\Rightarrow 4s \sqrt{t^2}$$

$$\Rightarrow 4s \cdot t$$

$$\boxed{\text{RHS} = 4 \sin\theta \cdot \tan\theta}$$

$$\text{LHS} = \text{RHS}$$

~~LP~~

Trigonometry समाप्ति!

Thank you Goodies !!

HW

Worksheet

